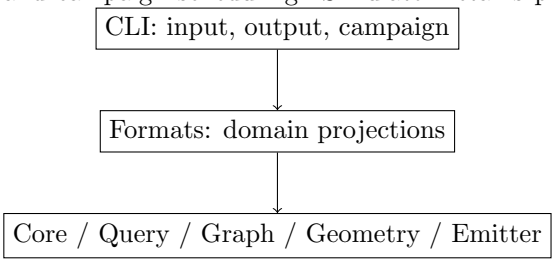
Research API boundaries

Created-at: 2026-10-07T15:43+02:00 (Europe/Budapest). ETA: 60 minutes for API extraction, focused regression and consumer command checks.

Scope Core remains compilation/IR/identity owner; Query supplies borrowed semantics; Graph owns search and spanning bases; Geometry owns FK/collision; Formats owns domain projections. Binaries own argv, output and campaign scheduling. Simulator retains physical execution. No new engine, crate or dependency version.



Files Formats examples adversarial_grid and four modules, metaheuristic_contract and profiles, mujoco_crosscheck; replacement src/environment modules, src/metaheuristic modules, src/robot_projection.rs and src/bin entries; src/lib.rs, src/codegen.rs, src/transforms.rs, Cargo.toml; Geometry src/transforms/mod.rs; integration tests; retired MJCF templates; Transformation README and shell launcher; Hystl research README; Simulator maze/mission docs if command changes. Generated SYMBOLS and this plan/report.

Interfaces Library APIs over Core snapshots; preserve schemas, caps, identities, deterministic priorities and output protection. Move existing Graph/Emitter dev dependencies to regular dependencies at unchanged revisions. Remove substantive examples. Route MJCF dispatch to its one existing articulated emitter.

Protected items None intended. No Simulator manifest or Kawamirai edits. No commits or pushes.

Tests Before/after existing tests; public API malformed/deterministic regression tests; CLI opt-in tests; scoped Clippy; dependency/duplication audit; Amaterasu fixed DE/GA and hex/polar artifact comparisons. Do not restart the overnight campaign.

Budgets and risks Preserve 128 KiB source, 4096 declarations, 16x16 grid, 1000 proposals, 8192-byte genome, 64 walls and 32 waypoints. Two compiler workers; no performance campaign. Risks: include paths, visibility, dropped checks, old executable paths, template bypass, unrelated dirty files. Worst input is existing bounded profiles, no scaling claim.

Rollback Pre-edit snapshots in ignored report artifacts; reverse only task hunks. Preserve historical run evidence and all unrelated work.

Final affected-file inventory Updated-at: 2026-10-07T16:11+02:00. The consumer audit also identified the existing template-dispatch integration test.

```
README.md
hymeko_formats/Cargo.toml
hymeko_formats/examples/adversarial_grid.rs
hymeko_formats/examples/adversarial_grid/maze.rs
hymeko_formats/examples/adversarial_grid/maze_physics.rs
hymeko_formats/examples/adversarial_grid/physical.rs
hymeko_formats/examples/adversarial_grid/visual.rs
hymeko_formats/examples/metaheuristic_contract.rs
hymeko_formats/examples/metaheuristic_contract/profiles.rs
hymeko_formats/examples/mujoco_crosscheck.rs
hymeko_formats/src/bin/adversarial_grid.rs
hymeko_formats/src/bin/adversarial_grid/maze.rs
```

hymeko_formats/src/bin/metaheuristic_contract.rs
hymeko_formats/src/bin/robot_projection.rs
hymeko_formats/src/codegen.rs
hymeko_formats/src/environment/maze.rs
hymeko_formats/src/environment/maze_physics.rs
hymeko_formats/src/environment/mod.rs
hymeko_formats/src/environment/physical.rs
hymeko_formats/src/environment/visual.rs
hymeko_formats/src/lib.rs
hymeko_formats/src/metaheuristic/mod.rs
hymeko_formats/src/metaheuristic/profiles.rs
hymeko_formats/src/robot_projection.rs
hymeko_formats/src/transforms.rs
hymeko_formats/tests/research_api.rs
hymeko_formats/tools/run_adversarial_grid.sh
hymeko_geometry/src/transforms/mod.rs
hymeko_query/tests/test_template_driven.rs
transforms/mjcf/queries.hymeko
transforms/mjcf/template.mjcf.xml